

Coding and Decoding (कोडिंग और डिकोडिंग)

Complete Theory, Solved Examples & Practice Set

1. Theory & Concepts (सिद्धांत)

Coding is a method of transmitting messages in a secret code or cipher so that no third person can understand it, and **Decoding** is the process to decrypt or understand the original message from the coded form.

To solve these questions efficiently, you must memorize the positional values of English alphabets (A=1 to Z=26) and their reverse pairs (A-Z, B-Y, etc.).

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Value	1	2	3	4	5	6	7	8	9	10	11	12	13
Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	14	15	16	17	18	19	20	21	22	23	24	25	26

Major Types of Coding-Decoding:

- **Letter Coding (अक्षर कोडिंग):** Forward or backward shifting of letters (e.g., +1, +2 or -1, -2).
- **Direct/Substitution Coding (प्रत्यक्ष कोडिंग):** Specific codes are assigned to specific letters or words directly.
- **Number/Symbol Coding (संख्या कोडिंग):** Alphabets are coded using their numerical positions or arithmetic calculations.
- **Matrix Coding (मैट्रिक्स कोडिंग):** Finding letters using row and column coordinates.
- **Conditional Coding:** Conditions given for vowels/consonants coding.

2. Solved Examples (उदाहरण और हल)

Example 1: If 'CAT' is written as 'DBU', how is 'DOG' written in that code?

Solution: Each letter is shifted forward by +1. $C(+1)=D$, $A(+1)=B$, $T(+1)=U$. Similarly, $D(+1)=E$, $O(+1)=P$, $G(+1)=H$. Answer: **EPH**

Example 2: If 'PEN' is coded as 'QFO', how is 'BOX' coded?

Solution: Pattern is +1 to each letter. $P \rightarrow Q$, $E \rightarrow F$, $N \rightarrow O$. For BOX: $B \rightarrow C$, $O \rightarrow P$, $X \rightarrow Y$. Answer: **CPY**

Example 3: If 'ROSE' is written as 'TQUG', how is 'BLUE' written?

Solution: Each letter is shifted by +2 ($R+2=T$, $O+2=Q$, $S+2=U$, $E+2=G$). For BLUE: $B+2=D$, $L+2=N$, $U+2=W$, $E+2=G$. Answer: **DNWG**

Example 4: If 'Z = 26' and 'NET = 39', then 'NUT' = ?

Solution: Sum of alphabetical positions. $NET = 14 + 5 + 20 = 39$. $NUT = 14 + 21 + 20 = 55$.

Example 5: In a code language, 'MANGO' is written as 'OCPIQ'. How is 'APPLE' written?

Solution: Pattern is +2 to each letter ($M+2=O$, $A+2=C$, etc.). For APPLE: $A+2=C$, $P+2=R$, $P+2=R$, $L+2=N$, $E+2=G$. Answer: **CRRNG**

Example 6: If 'SISTER' is coded as '535901' and 'UNCLE' as '84670', how is 'SON' coded?

Solution: Direct substitution from given words. $S=5$, $O=9$, $N=0$. Answer: **590**

Example 7: If 'GOOD' is written as 'HPPE', how is 'BAD' written?

Solution: Each letter is shifted by +1. $G \rightarrow H$, $O \rightarrow P$, $O \rightarrow P$, $D \rightarrow E$. For BAD: $B \rightarrow C$, $A \rightarrow B$, $D \rightarrow E$. Answer: **CBE**

Example 8: If 'WATER' is written as 'XBUFS', how is 'FIRE' written?

Solution: Pattern is +1 for each letter. W→X, A→B, T→U, E→F, R→S. For FIRE: F→G, I→J, R→S, E→F. Answer: **GJSF**

Example 9: If 'BOOK' is coded as '43', what is the code for 'PEN'?

Solution: Sum of positional values: B(2) + O(15) + O(15) + K(11) = 43. For PEN: P(16) + E(5) + N(14) = **35**.

Example 10: If 'RED' is coded as '6720', how is 'GREEN' coded?

Solution: Reverse alphabetical positions used or positional value multiplied/formatted. Here, reverse positions: R(9), E(22), D(23). Alternatively, R=18→18+2=20, E=5→5+2=7, D=4→4+2=6 (written in reverse: 6, 7, 20). For GREEN (reverse pattern): N=13+2=15, E=5+2=7, E=7, R=18+2=20, G=7+2=9. Answer: **7222209** (or standard positional reverse logic). Let's use simple shift reverse: Answer: **VIVTII** (using opposite letters: R-I, E-V, D-W). Standard opposite: R→I(9), E→V(22), D→W(23). Code = 23,22,9.

Example 11: If 'STUDY' is coded as 'TVVEZ', how is 'LEARN' coded?

Solution: Pattern is +1, +2, +3, +4, +5 respectively. L(+1)=M, E(+2)=G, A(+3)=D, R(+4)=V, N(+5)=S. Answer: **MGDVS**

Example 12: If 'MUMBAI' is written as 'LTLAZH', how is 'DELHI' written?

Solution: Each letter is shifted backward by -1. M(-1)=L, U(-1)=T, etc. For DELHI: D(-1)=C, E(-1)=D, L(-1)=K, H(-1)=G, I(-1)=H. Answer: **CDKGH**

Example 13: If 'A = 1' and 'AND = 19', then 'BAT' = ?

Solution: AND = 1 + 14 + 4 = 19. BAT = B(2) + A(1) + T(20) = **23**.

Example 14: If 'FORCE' is written as 'GPSDF', how is 'GREEN' written?

Solution: Pattern is +1 to each letter. F→G, O→P, R→S, C→D, E→F. For GREEN: G→H, R→S, E→F, E→F, N→O. Answer: **HSFFO**

Example 15: If 'TABLE' is written as 'UZCQG', how is 'CHAIR' written?

Solution: Pattern is +1, +2, +3, +4, +5. C(+1)=D, H(+2)=J, A(+3)=D, I(+4)=M, R(+5)=W. Answer: **DJDMW**

3. Practice Questions (अभ्यास प्रश्न)

Q1. If 'FISH' is written as 'GJTI', how is 'MAN' written?

- (A) NBO (B) MBO (C) NCQ (D) OBP

Q2. If 'APPLE' is written as 'ETPPI', how is 'ORANGE' written?

- (A) SVERKI (B) SWFRKJ (C) RUFOKI (D) TVGSLK

Q3. If 'DOG' = 26, what is 'CAT' = ?

- (A) 20 (B) 24 (C) 22 (D) 25

Q4. If 'ROHA' is coded as 'SPIB', how is 'DELHI' coded?

- (A) EFMIJ (B) EGMIJ (C) EFMHJ (D) DEMIH

Q5. If 'ROAD' is written as 'URDG', how is 'SWAN' written?

- (A) VZDQ (B) VZCD (C) UZDQ (D) VZDQ

Q6. If 'Z = 52' and 'ACT = 48', then 'BAT' = ?

- (A) 46 (B) 50 (C) 39 (D) 44

Q7. If 'MONKEY' is written as 'XEDNML', how is 'TIGER' written?

- (A) QDFHS (B) SDFHS (C) QDHFS (D) REFIT

Q8. If 'PEN' = 35, then 'PAGE' = ?

- (A) 22 (B) 20 (C) 24 (D) 28

Q9. If 'STUDENT' is written as 'TUTEFOU', how is 'TEACHER' written?

- (A) UFBBIFS (B) UFBBFIS (C) UFBGJFS (D) VECCJGT

Q10. If 'MADRAS' is written as 'NBESBT', how is 'BOMBAY' written?

- (A) CPNCBZ (B) CPNCBX (C) BQOCAZ (D) CQPCBY

Q11. If 'NUMBER' is written as 'UNBMRE', how is 'GHOSTS' written?

- (A) HGSTSO (B) HOGSST (C) OHGSTS (D) HGSSOT

Q12. If 'B = 2' and 'BAG = 10', then 'BOX' = ?

- (A) 39 (B) 41 (C) 43 (D) 45

Q13. If 'WHITE' is written as 'XKJYM', how is 'BLACK' written?

- (A) CMDFP (B) CNEGP (C) CLBDO (D) CNDHP

Q14. If 'CIRCLE' is written as 'DKSLOJ', how is 'SQUARE' written?

- (A) TRVCSX (B) TRWCXG (C) TRVDTI (D) TRWEXI

Q15. If 'EARTH' is written as 'FCVWM', how is 'MOON' written?

- (A) NRRV (B) NQRT (C) NQSU (D) NRRU

Q16. If 'JAM' = 26, 'KITE' = 42, then 'BOY' = ?

- (A) 35 (B) 38 (C) 42 (D) 45

Q17. If 'SHIMLA' is written as 'RGHLKZ', how is 'PUNE' written?

- (A) OTMZ (B) QTLN (C) OTMD (D) NTMD

Q18. If 'PAPER' is written as 'SDRHU', how is 'PENCIL' written?

- (A) SHQFLO (B) SHQFLP (C) SGQFLO (D) RHQFLO

Q19. If 'WATER' is written as 'YCVGT', how is 'CHAIR' written?

- (A) EJKKT (B) EJCKT (C) EJCKU (D) DJCKT

Q20. If 'GARDEN' is written as 'HBSEFO', how is 'FLOWER' written?

- (A) GMPXFS (B) GMPXFT (C) GMQXFS (D) HMPXFS

Q21. If 'SOLID' is written as 'WPPMH', how is 'BRING' written?

- (A) FVMQK (B) FVMQJ (C) EVMQK (D) FULQK

Q22. If 'DELHI' is written as 'CCKGD', how is 'BOMBAY' written?

- (A) ANLAXW (B) ANLAXV (C) BNLAXW (D) AMKAXW

Q23. If 'HOT' = 45, then 'COLD' = ?

- (A) 27 (B) 32 (C) 35 (D) 38

Q24. If 'TRAIN' is written as 'RNYGL', how is 'SCOOTER' written?

- (A) QAMMRCP (B) QAMMRBP (C) QAMMRCQ (D) PAMMRCP

Q25. If 'STUDY' is written as 'UTWFA', how is 'LEARN' written?

- (A) NGCTP (B) NGCSP (C) NGCVP (D) NGCTQ

4. Answer Key (उत्तरमाला)

Q1. (A)	Q2. (A)	Q3. (C)	Q4. (A)	Q5. (A)
Q6. (D)	Q7. (A)	Q8. (A)	Q9. (A)	Q10. (A)
Q11. (B)	Q12. (B)	Q13. (A)	Q14. (B)	Q15. (A)
Q16. (B)	Q17. (C)	Q18. (A)	Q19. (B)	Q20. (A)
Q21. (A)	Q22. (A)	Q23. (B)	Q24. (A)	Q25. (A)